MoleCheck

Privacy-Preserving Skin-Lesion Analysis

Capstone Project Report

A mobile application that classifies a photographed skin lesion as lower- or higher-risk using a convolutional neural network, with all inference running on-device — no server, no upload, full offline operation.

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1. Summary

MoleCheck classifies a photograph of a single skin lesion as lower-risk (benign) or higher-risk (malignant) using a convolutional neural network. Inference runs **entirely on-device**: the model is bundled inside the application, executes on the phone's own hardware, and functions in airplane mode. Because the image never leaves the device, the design provides a strong privacy guarantee for sensitive medical imagery.

Evaluated on a held-out test set of 1,722 images, the model achieves a **ROC-AUC of 0.914**, detecting **90% of malignant lesions (sensitivity)** while correctly clearing **75% of benign lesions (specificity)**.

> Not a medical device. MoleCheck is an educational demonstration of image classification. It cannot diagnose skin cancer or any other condition and must not inform a health decision. Anyone concerned about a mole or a skin change should consult a dermatologist.

2. Motivation

Skin cancer is among the most curable cancers when detected early and among the most dangerous when it is not. Access to dermatological screening is uneven, and most people cannot judge which lesions, if any, warrant professional review. The goal of this project was to build a complete pipeline end to end: train a classifier on real dermatological images, export it to run on a phone with no server, and deliver it inside a functioning mobile application. The project therefore spans machine learning, model deployment, and mobile development.

3. Dataset and labeling

Training used a public ISIC Archive collection of **11,720 dermatoscopic images** (a superset of the HAM10000 dataset). Each image's diagnosis field was mapped to a binary label — benign or malignant — with indeterminate cases excluded. The dataset is heavily imbalanced toward benign lesions, which informed both the training procedure and the choice of decision threshold.

Preventing data leakage

The same lesion is sometimes photographed more than once. Allowing multiple images of one lesion to fall on both sides of the split would let the model memorize specific lesions and report inflated performance. To prevent this, the data was partitioned so that **every image of a given lesion remains in a single split**.

Split	Lesions	Images	Malignant %
Training	6,121	8,123	18.8%
Validation	1,309	1,726	18.1%
Test	1,309	1,722	18.2%

4. Model and training

The classifier is **YOLO11s-cls**, the classification variant of Ultralytics' YOLO11 (5.4M parameters), fine-tuned for 40 epochs on an NVIDIA RTX 5060 Ti via **transfer learning** from a pretrained backbone. A pretrained model with a clean mobile-export path was chosen deliberately: a strong, reproducible baseline that deploys reliably is more valuable here than a bespoke architecture that is difficult to ship. Standard augmentation (flips, rotations, and color jitter) was applied so the model is robust to variation in lighting and orientation.

5. Results and threshold selection

Accuracy is a misleading metric on this task: because most lesions are benign, a trivial "always benign" classifier already attains roughly **82%** accuracy while detecting no cancers. The evaluation therefore emphasizes sensitivity (malignant lesions detected) and specificity (benign lesions correctly cleared).

Metric	Value	Interpretation
ROC-AUC	0.914	Overall separability of the two classes
Sensitivity	90.1%	Malignant lesions correctly flagged
Specificity	74.7%	Benign lesions correctly cleared

Prioritizing sensitivity

The default 0.5 decision threshold is inappropriate for a screening task, because the two error types are asymmetric: a false negative (a malignant lesion reported as benign) is far more costly than a false positive (a benign lesion referred for review). The operating threshold was set to **0.137**, the point at which the model detects 90% of malignant lesions while still clearing 75% of benign ones. The application is deliberately tuned to err toward recommending professional evaluation.

6. Deployment and verification

On-device inference required converting the PyTorch model to TensorFlow Lite. Because the direct export path is unsupported on Windows, the model was routed through ONNX and converted with onnx2tf. Conversion correctness was **verified rather than assumed**: the exported model was evaluated against the original PyTorch model and reproduced its predictions to within 0.001 probability, with identical decisions. The model was then integrated into a **Flutter** application (a single codebase targeting Android and iOS), gated behind a first-run disclaimer acknowledgement, and tested on an Android device.

7. Limitations

- Not a medical device — an educational project, not validated for clinical use.
- A 90% sensitivity target still implies roughly one in ten malignant lesions is missed, so a 'lower-risk' result is not reassurance.

- The model was trained on dermatoscopic images, which differ substantially from ordinary smartphone photographs; real-world accuracy is expected to be lower.
- Public dermatology datasets under-represent darker skin tones, so performance is not guaranteed to be uniform across skin types — a recognized equity concern in dermatology AI.

These limitations are precisely why the application always directs the user to consult a dermatologist.

8. Future work

- Train and evaluate on ordinary smartphone photographs, not only dermatoscopic images.
- Measure and address performance disparities across skin tones.
- Add an abstention option so low-quality images are declined rather than classified.
- Complete the iOS build and conduct informal user testing.